

SUMMARY TABLE: MATHEMATICS GRADE 10

Sub-Strand 3.2: Rotation — Teacher Reference (Pre-filled)

SUMMARY TABLE: MATHEMATICS GRADE 10	
Sub-Strand	Sub-Strand 3.2: Rotation
Driving Question	DRIVING QUESTION: Why do the blades of the Ngong Hills wind turbine always look the same no matter how far they rotate — and how can we use the mathematics of rotation to predict exactly where any point on the blade will land?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Anchoring Phenomenon & Introduction to Rotation	Students watched the Ngong Hills Wind Farm turbine video and recorded observations in a three-column table (what is moving, what stays the same, what is puzzling). They noted that individual blades move through space but the overall three-blade assembly appears unchanged at every stop position, and that the central hub point never moves.	Students were introduced to the vocabulary of rotation: centre of rotation (the fixed point — the turbine hub), angle of rotation (how far a point moves around the centre), direction of rotation (clockwise = negative; anticlockwise = positive), object (original position) and image (new position). They co-constructed a working definition: rotation moves every point on a shape through the same angle around a fixed centre, preserving shape and size.	Students began to explain the anchoring phenomenon: the Ngong Hills turbine blades always look the same because the three-blade assembly is being rotated about its central hub, and rotation preserves the shape and size of the figure. However, students cannot yet explain why exactly 120° produces an identical-looking assembly, how to calculate exactly where any point on a blade will land, or how to determine the centre and angle of rotation from a diagram alone — these questions are recorded on the DQB and will drive Lessons 2–6.
Lesson 2 Properties of Rotation: What Stays the Same and What Changes?	Learners rotated cardboard cut-out shapes (modelled on the Ngong Hills turbine blades) about a fixed pin on grid paper. They measured the distances from the centre O to each vertex of the object and to the corresponding vertex of the image for rotations of $+90^\circ$ (anti-clockwise) and -120° (clockwise), and recorded all values in an evidence table. They also performed a congruence check by overlaying the object and image cut-outs.	Rotation has four defining properties: (1) the centre of rotation O is fixed and does not move; (2) the distance from O to any point on the object equals the distance from O to its image — $OA = OA'$ (distance is invariant); (3) the object and its image are always congruent — same shape, same size, only position changes; (4) direction matters — a positive angle means anti-clockwise rotation and a negative angle means clockwise rotation. The angle of	These four properties explain why the Ngong Hills wind turbine blades always look the same no matter where they stop: the hub is the fixed centre of rotation, each blade tip is the same distance from the hub before and after turning (distance invariant), and the entire three-blade assembly is congruent to itself after every rotation — the blades have simply changed position, not shape or size. The direction of spin (positive = anti-clockwise when viewed from the

		rotation is measured at the centre, from the line joining O to the object point to the line joining O to the image point.	front) is a mathematical fact about the rotation that engineers can measure and predict precisely.
Lesson 3 Rotating Points on the Cartesian Plane	Students watched a simulation and used Cartesian grids to physically rotate points A(3,1) and B(-2,3) through 90°, 180°, and 270° anticlockwise about the origin, recording image coordinates in a data table and comparing hand-plotted results with the digital simulation output.	Students derived and applied the four coordinate rotation rules about the origin: 90° anticlockwise $(x,y) \rightarrow (-y,x)$; 90° clockwise / 270° anticlockwise $(x,y) \rightarrow (y,-x)$; 180° $(x,y) \rightarrow (-x,-y)$; and confirmed that a positive rotation angle means anticlockwise while a negative angle means clockwise.	Students explained that the Ngong Hills turbine blade tip at P(4,2) moves to predictable, calculable positions — (-2,4), (-4,-2), and (2,-4) — after successive 90° anticlockwise rotations, and that after four such rotations it returns exactly to (4,2), beginning to explain mathematically why the blade always looks the same regardless of its rotational position.
Lesson 4 Rotating Shapes on the Cartesian Plane	Students observed that when every vertex of triangle KMN was rotated 90° anticlockwise about the origin, the resulting image triangle K'M'N' had exactly the same side lengths and angles as the original — confirmed by measurement with a ruler and by GeoGebra. They also observed that rotating quadrilateral ABCD by 180° about the origin produced an image that was congruent to the object and positioned symmetrically about the origin.	Students learned the coordinate transformation rules for rotation about the origin: 90° anticlockwise maps (x, y) to $(-y, x)$; 180° maps (x, y) to $(-x, -y)$; 90° clockwise maps (x, y) to $(y, -x)$. They also learned that rotation about a non-origin centre requires each vertex to be rotated through the same angle about that specific centre point, and that positive angles indicate anticlockwise direction while negative angles indicate clockwise direction.	Students explained that the Ngong Hills turbine blades always look the same no matter where they stop because rotation is a congruence-preserving transformation — every point on the blade moves through the same angle about the fixed hub, so the overall shape and size of the blade is unchanged. The three blades, spaced 120° apart, each map exactly onto the next blade's position after a one-third turn, which is why the turbine assembly appears identical at every stopping point.
Lesson 5 Angle and Centre of Rotation — Finding What We Cannot See	Learners observed a teacher-led compass-and-ruler construction showing how perpendicular bisectors of segments joining object-image point pairs (PP' and QQ') intersect at a single point — the centre of rotation. They also read an explanation of WHY this works: because rotation preserves distance from the centre, every object-image pair is equidistant from the centre, so the centre must lie on the perpendicular bisector of each such segment.	Learners learned that: (1) the centre of rotation is the unique point equidistant from every object point and its corresponding image point; (2) constructing perpendicular bisectors of two object-image point segments and finding their intersection locates the centre; (3) the angle of rotation is measured at the centre O between the ray to the object point and the ray to the image point; (4) a positive (anti-clockwise) or negative (clockwise) sign must accompany the angle. They also learned that this construction can reverse-engineer a rotation — given only two blade positions, the turbine hub's location is fully determined.	Learners explained how — given only the positions of a Ngong Hills turbine blade before and after a rotation — they can construct perpendicular bisectors of two corresponding vertex segments to locate the exact centre of rotation (the turbine hub), and then use a protractor at that centre to measure and state the angle and direction of rotation. They updated their cumulative model and DQB to reflect that the hub is not merely a visible physical point but a geometrically necessary and recoverable centre of rotational symmetry.
Lesson 6 Order of Rotational Symmetry	Students watched a video of the Ngong Hills three-bladed wind turbine and animated regular polygons (equilateral triangle, square, regular pentagon, regular	A shape has rotational symmetry of order n if it maps exactly onto itself n times during a full 360° rotation. The minimum angle of rotation that produces a self-mapping is	The three-bladed Ngong Hills turbine always looks the same at certain rotation positions because the three blades are equally spaced at 120° apart, giving the

	hexagon) rotating about their centres, as well as a Maasai beadwork 6-petal rosette and a square Kanga tile unit. They recorded in an observation table every angle at which each shape mapped onto itself within one full 360° turn, and counted the number of times each shape did so.	calculated as $360^\circ \div n$. The Ngong Hills three-bladed turbine has rotational symmetry of order 3, with self-mappings at 120°, 240°, and 360°. This is why the blade assembly always looks the same at those positions — the rotation maps every blade onto the position of another blade.	assembly a rotational symmetry of order 3. Every time the turbine rotates 120° anticlockwise (positive rotation), each blade moves into the exact position the previous blade occupied, so the overall shape is unchanged. This is a property called rotational symmetry, and the order tells us how many times this identical mapping occurs in one full 360° turn.
Lesson 7 Axis and Order of Rotational Symmetry: From 2D Figures to 3D Objects	Students observed that 3D objects such as a sufuria, a bottle cap, a gear, and the Ngong Hills turbine hub each rotate about a fixed line called an axis of rotational symmetry, and that they map onto themselves a countable number of times in one full rotation.	The axis of rotational symmetry is the fixed straight line about which a 3D object maps onto itself during rotation. The order of rotational symmetry and minimum angle formula (360 degrees divided by order) apply in exactly the same way in 3D as in 2D. The turbine rotates about the horizontal shaft through its hub — this is its axis — and has order 3 with minimum angle 120 degrees.	The reason the Ngong Hills turbine blades always look the same can now be described with full 3D precision: the assembly has rotational symmetry of order 3 about the horizontal axis through the hub, meaning every 120 degrees of rotation maps the blade assembly exactly onto itself. This is not just a flat geometric property — it is a physical property of the real machine spinning in three-dimensional space.
Lesson 8 Congruence and Rotation: Proving the Turbine Blades Are Congruent	Students observed (through direct measurement on worksheets) that when a shape is rotated, the side lengths and angle sizes of the image are identical to those of the original object. A video of a Maasai shield motif being rotated also showed on-screen measurements remaining unchanged after rotation.	Students learned that rotation is an isometry — a transformation that preserves length, angle size, and area — making the image congruent to the original. They learned that rotation, reflection, and translation are all congruence transformations, but only reflection reverses orientation. They formalised that the three Ngong Hills turbine blades are congruent because each is the image of the previous blade under a 120° anticlockwise rotation about the central hub.	Students explained that the turbine blades always look the same because rotation is an isometry: no matter how far the blades spin, their shape and size cannot change. Each blade is mapped perfectly onto the next by a 120° rotation, proving the three blades are congruent — which is exactly why the turbine appears visually identical in every rotational position.
Lesson 9 Applications of Rotation in Real-Life Situations	Learners observed four real-life Kenyan rotation contexts at carousel stations: a rotationally symmetric logo design, analogue clock faces showing swept angles, an illustrated model of the Earth rotating on its axis to create day and night in Nairobi, and a meshed gear diagram showing the relationship between driver and driven gear rotations.	Learners learned that: (1) the angle swept by a clock's minute hand is 6° per minute ($360^\circ \div 60 \text{ min}$), making it a negative (clockwise) rotation; (2) the Earth rotates 15° per hour on its axis, and Nairobi must rotate 180° from midday to midnight; (3) gear rotation angles are inversely proportional to the number of teeth (a 24-tooth gear driving an 8-tooth gear produces a 3:1 rotation ratio); (4) rotationally symmetric logos are produced by rotating a base shape through equal angles (e.g., 120° for order 3, 90° for order 4) about a	Learners explained that all rotating systems — the Ngong Hills turbine, clock hands, the Earth, and gears — are governed by the same rotation mathematics: a fixed centre of rotation, a measurable angle with a direction (positive = anticlockwise, negative = clockwise), and the preservation of distance from the centre. The turbine blades 'always look the same' because each 120° rotation maps the blade assembly perfectly onto itself — a property shared by every rotationally symmetric object, from Kenyan logos to the Earth's

		fixed centre — the same mathematics that governs the Ngong Hills turbine blades.	daily spin.
Lesson 10 Unit Consolidation, Model Finalisation & DQB Review — Answering the Driving Question	Students watched the Ngong Hills Wind Farm turbine video again (the same clip from Lesson 1) and observed that the three equally-spaced blades, each separated by 120°, cause the entire blade assembly to map perfectly onto itself three times in one full 360° rotation. They also observed, through the evidence gallery walk, a consolidated display of all unit evidence: direction conventions (positive = anti-clockwise, negative = clockwise), fractional turns ($90^\circ = \frac{1}{4}$ turn, $120^\circ = \frac{1}{3}$ turn), centre-of-rotation construction, and congruence preservation.	Students learned that the driving question — 'Why do the blades of the Ngong Hills wind turbine always look the same no matter how far they rotate?' — is fully answered by five interconnected mathematical ideas: (1) Rotation is defined by a centre, an angle, and a direction; (2) A positive angle means anti-clockwise rotation, a negative angle means clockwise rotation; (3) The three blades are equally spaced at 120° intervals, giving the assembly an order-3 rotational symmetry — meaning it maps onto itself 3 times in 360°; (4) Rotation is an isometric transformation, so every blade image is congruent to the original — lengths, angles, and area are all preserved; (5) Given any object–image pair, the centre and angle of rotation can be determined precisely using perpendicular bisectors and angle measurement.	Students explained the driving question in their own written Final Explanations (Q5 of the summative task) using all five required mathematical terms. The class-level consensus explanation is: 'The blades of the Ngong Hills wind turbine always look the same because the three blades are equally spaced at 120° from one another around the fixed centre of rotation at the hub. When the turbine rotates by 120° anti-clockwise (a positive rotation), the entire blade assembly maps perfectly onto itself — this is order-3 rotational symmetry. Because rotation is an isometric transformation, every blade in its new position is congruent to its original position: all lengths, angles, and areas are preserved. This means no matter how many 120° turns the turbine makes — whether clockwise (negative angles) or anti-clockwise (positive angles) — the shape always appears identical. Using the mathematics of rotation, we can predict exactly where any point on a blade will land after any given angle of rotation about the hub.'

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